TITLE: Phan Tuấn Kiệt 2352654 Date: HW1

Activity 1:

according to VietNam post corporation, an envelope must contain the following information for the system can sort correctly:

Recipient information:

+ full name of reciv. Recipient
+ Recipient's address
+ Zip code of Recipient's area

Sender information:

+ Sender's fullname
+ Sender's address
+ Sender's phone number

The target of the envelope sorting system is to determine the destination area for each envelope.

Some ~~decision~~ predict on ~~[illegible]~~ received from the system are: the destination of envelops; wrong information; need confirm from sender; Unable to verify the information.

For example:

- if sender writes ~~wrong~~ a address that not exist in this area → wrong information
- system cannot read ~~the~~ address but can detect the sender's phone number → call back to sender
- the empty information envelop ~~or the~~ → unable to verify information.

There are some noise that can affect to ~~the~~ prediction of system

- ~~the~~ ink color used for writing information.
- envelop's size
- bad handwriting
- cross-out mark
- ...

Depending the goal of system, developer can use this raw data for build a program that classify the destination of the envelope Or use ~~the~~ detected information TO predict the lack of information like zip code.

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Activity 2:

(I) 80/20: 80% of samples use for training and 20% samples for validation set, which mean that 96 samples for training set and 24 samples for validation set. This methods trains model only one time.

We don't know if the data distribution evenly between the training ~~set~~ set and the validation set. Specifically, if the distribution of feature A in traing set more higher than ~~class B~~ feature B while validation set has a higher percentage of featur B compared to feature A, this situation can affect to the model because it may optimize for validation set.

II 5 fold: the data set split into 5 parts equally, each fold have 24 samples the training process repeat 5 times, each time use the different fold as validation set and the remain folds as training set.

With small samples like this dataset, this method may helpful because it make sure that each sample have one time to become ~~validation set.~~ an element of validation set.

The reason why using the same test set while chooing hyperparameter can make the final test result optimistic:

The goal of change hyperparameter is help the model become more accurate, but use the same test set make the model tend to satisfy the test set, leading to phenomenon of overfitting.

Activity 3:

$$\text{Accuracy} = \frac{TN+TP}{TN+TP+FN+FP} = \frac{170+36}{170+36+14+30} = 0,824$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{36}{36+30} \approx 0,5455$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{36}{36+14} = 0,72$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{0,5455 \times 0,72}{0,5455+0,72} \approx 0,6207$$

This model use for detect the rare event, if use accuracy ~~to~~ to evaluate it can be misleading, as the value is 82,4% which is quite good result. In reality, the model can only accuracy identify the target event 62,07% of the time.

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Activity 4.

A = (24%, 26%) training error and validation error have nearly the same value and high. After training and validation, the model have more error for both set mean that this model unable to cover the data distribution regions. → underfitting

The solution is to make the model mor complex.

B = (3%, 21%) training error is small but validation error ~~high~~ much higher that training error. The model do very well for training set but when apply it for validation set, it not active good. This model can be concluded overfitting because it just do well for training set.

The solution is ~~to~~ make the model simpler.

C = (9% ÷ 11%) compare to model B, the accuracy of training set is lower than model B but it have ~~the~~ higher accuracy of validation set. ~~The~~ Both sets have nearly the same ~~and small enough~~ error rate and small enough ~~to~~. → this model is a good model.

Activity 5.

My data set have 6 class and 500+ features. and 7300 samples.

~~7300 samples~~ is My dataset ~~have~~ also have a~~nother set~~ isolate test set

So I will split 7300 samples randomly into 80% for training set and 20% for validation set.

My model ~~[illegible]~~ uses for predicting the user's current physical activity based on ~~[illegible]~~ internal sensor from smartphone.

There are 6 lables: walking, walking upstairs, walking downstairs, sitting, standing, laying

6 events ~~have the same~~ are equal so I use macro - F1 score to evaluate model.

To make easy for compare the algorithm performance. First of all I random split the data set into 80% for training set and 20% for validation set, Then I save these splited dataset and use it for all requirement from this course ~~[illegible]~~.

Just write it

Because I split data randomly, it can happen the situation that have a label that just only have in a set (training or validation set), if this situation occur, I will split again until ~~do~~ this situation disappear.

There ~~[illegible]~~ a limitation that cannot remove ~~[illegible]~~ If the validation set have ~~[illegible]~~ most of easy ~~values~~ Input which ~~[illegible]~~ are not have enough challenge for training set leading to the result may be high abnormally.